Six-obligation follow-up: first attacks and hard gates

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Status: two implementation advances validated; four theory/measurement lanes sharpened to checkable next objects; no scientific performance run taken.

Outcome

This continuation did not treat the six priorities as six interchangeable benchmarks. It separated source-theorem admission, correctness proof kernels, scientific panels, and mathematical lower-bound searches. The current state is:

- Tensor: exact BigUint D.4 guard mirror and first-boundary regression; predicate cost and aggregate complexity remain uncertified.
- Yield/rank: the named batch's off-main custody hold was verified before execution; `BATCH-79a580` may not be rerun, reclaimed, or silently adopted.
- Arity three: source-bound unit-equation explanations and independent replay; this is not yet a CaDiCaL propagator.
- PKM: the closure-dual target is now an exact nullspace certificate; no growing certificate has been found.
- Wagner: the exact `W` statistic, estimator gates, nulls, and both decision directions are recovered; a native frozen producer is missing.
- Xedni: selector quantifiers and a model-rescaling falsifier are explicit; no selected-height/global-minimal-discriminant theorem is proved.

The dependency and gate view is in `obligation-attack-graph.svg`. The rendered report is `FOLLOWUP.pdf`.

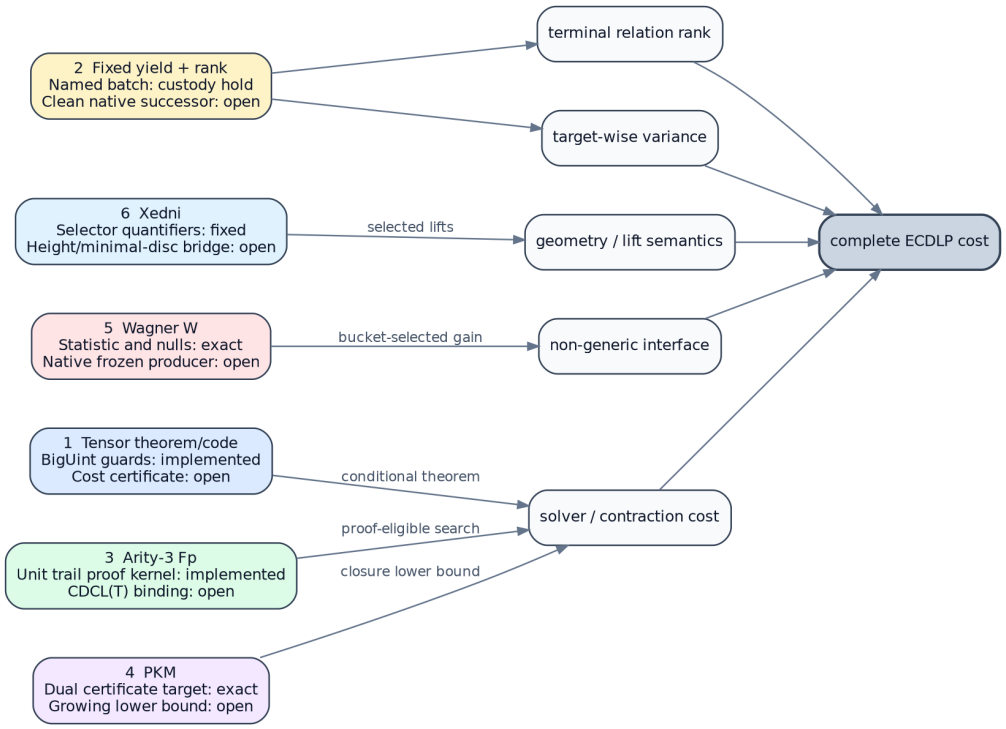


Figure 1: Current attack graph: blue/green nodes have a new implementation artifact; yellow/purple/red nodes are sharpened but still blocked.

1. Tensor contraction theorem-to-code audit

The audit used OpenAI Appendix D at commit `7ef5c0cc...` and the Bochum presentation at `1ee44e56...`. The important correction is that the recursive kernel is not an unguarded transcription. It has explicit runtime checks for most structural invariants. The remaining gap is concentrated in theorem admission and complexity accounting.

- **Graph and degree.** `Bipartite::from_boolean` performs the stated two-edge subdivision. `Setup::new` computes `ceil(s^(9/10))` exactly and forces fallback when the degree hypothesis fails. Both are mapped and fail closed.
- **D.4 parameters and guards.** The implementation uses the source `lambda`, leaf cutoff, epsilon `1/(128*1024)`, radix 64, `u_d`, `a_d`, `L`, and `D`. Variance, packing, and `2L<=Delta` are checked before recursion.
- **Packing.** Conditional expectation minimizes an integer second-moment objective. Every old edge is checked exactly once; packet count, padding, and each coordinate's group port balance are checked.
- **Modes and children.** Source-group orientation is checked while building incoming/outgoing forms. Strict children check eightfold edge shrink and the degree cap. Bounded mode tests exist; a general proof remains open.
- **Implicit tensors.** Packet descriptions store local matrices and forms, with no parent evaluator callback. Entries are evaluated at immutable creation precision and only then lifted.
- **Recovery.** Node precision, jet shape, grid coverage, denominators, and return to the base coefficient ring are checked. These are bounded controls, not a general formal proof.
- **Open admission gap.** Canonical IR provides a finite typed representation, but public `FnPredicate` may be opaque. Neither polynomial description size nor the uniform `exp(A*s^(9/10))` entry-cost hypothesis is enforceable for such a closure. Meter units are not a theorem cost model.
- **Open aggregate gap.** Local allocations are bounded and recursion is depth-first, but no source-level aggregate construction/space proof is bound to an admitted instance family.
- **Representation boundary.** Live setup is `usize`-indexed. This prevents an executable asymptotic branch but is not a counterexample to the theorem.

The new public module `admission.rs` mirrors the numeric source guards with `BigUint`. The CLI `tensor-admission-audit [SIZE|2^K+N] [MAX_DEGREE] [DEPTH]` reports only numeric admission and explicitly refuses to certify predicate or aggregate costs.

At the root, `u=floor(lambda/131072)`. The first even subdivided edge count with nonzero `u` is therefore

```
s0 = 2^131071 + 2.
```

At `s0`, degree two, and depth zero, the exact evaluator obtains `lambda=131072`, `u=1`, `a=2`, `L=4`, `D=2^24`; every numeric D.4 guard passes, while `s0` is not live-host representable. Four Rust regressions pass, including the exact boundary and the 64-bit zero-scale statement. This establishes a precise interface fact: practical public calls use exact fallback, while the source recursion begins beyond the graph representation. It does not refute the asymptotic theorem and does not validate a performance claim.

The next tensor step is not another small forced-recursion timing. It is a typed admission object constructed from canonical network bytes that gives a checkable description-size and per-operation upper bound for every predicate family, followed by an aggregate construction/space proof. Arbitrary closures must remain ineligible.

2. Fixed-budget yield-count-vector plus rank

The atlas's requested panel is not ready to dispatch. Its own problem record, `reln-f2-p5-off-main-custody-unreconciled.md`, requires zero scientific runs and forbids rederivation, reclaim, or receipt rewriting until source ownership and native adoption are reconciled. The eight retained source blobs hash-match their historical receipt, but the source commit is not an ancestor of inspected main. Hash equality is custody evidence, not scientific admission.

The existing native `relation_yield_pruning` analyzer is also insufficient for this question. It accepts episode aggregates (`relation_hit_trials`, verified rows, rank, coverage, entropy), not the full per-target decomposition-count vector or immutable sparse row population. It cannot reconstruct zero mass, Fano factor, multiplicities, incremental rank, or graph nulls after the fact.

A clean successor must therefore emit, before any adaptive stop:

1. a fixed ordered natural-target vector and exact probe budget;
2. the exact count `c_D` for every target, including zeros and solver failures;
3. every independently replayed relation row with target and multiplicity;
4. rank after each row in the fixed order, dead columns, support components, 2-core, and terminal deficiency;
5. a size-matched random-base arm and rewired/Erdos–Renyi graph nulls; and
6. source, fixture, binary, protocol, checkpoint, and output hashes.

The yield and rank analyses must consume the same write-once population. The producer may report an operational recovery point, but collection must continue to the frozen budget; otherwise the count-vector distribution is selection biased. A new successor namespace may be built natively after its fixtures are publicly derived. It must not inherit the historical batch identity.

3. Proof-eligible arity-three finite-field propagation

The canonical quadratic IR already binds every equation, variable role, field, curve, target, and projected variable. The new `trail.rs` adds a small independently replayable theory-proof kernel:

- premises are strictly ordered canonical field assignments;
- the explanation names one exact source equation and the canonical IR hash;
- substitution must leave either a nonzero constant (conflict) or exactly `a*x+b` with `a!=0` (unique propagation);
- nonlinear residuals, two remaining unknowns, satisfied equations, unknown variables, noncanonical residues, changed hashes, and changed conclusions fail closed.

Three Rust tests pass: an affine `target_zero` propagation, a constant conflict, and rejection of a nonlinear/underassigned residual. A mutated propagated value is rejected. This is the first proof-eligible finite-field trail rule in this study, but it is intentionally not described as the production propagator.

The remaining implementation order is now concrete:

1. translate a checked field consequence into the Boolean selector literals that formed its premises and conclusion;
2. attach these clauses through CaDiCaL's external-propagator interface;
3. replay every emitted clause with this independent kernel plus the Boolean proof log;
4. add checked certificate types for source-equation linear combinations and guarded inversions, one rule at a time; and
5. only then freeze the `s=6..14` natural/planted/UNSAT panel.

The current kernel may be too weak to improve search. That is an empirical outcome to measure later, not a reason to weaken proof eligibility.

4. PKM pseudo-solution and matrix width

The most direct lower-bound target can be stated as a finite certificate. For degree `c`, construct the HKY closure space `V_{F,c}` in the tower normal-form basis. A degree-`c` pseudo-solution is a linear functional `L` satisfying

```
L(v) = 0 for every generated closure row v in V_{F,c},
L(1) = 1.
```

Equivalently, if `A_c` is the exact closure matrix with the constant column distinguished, find `ell` with `A_c * ell = 0` and `ell_constant=1`. Direct matrix multiplication verifies the certificate and proves `1` is not in the closure, hence `d_F>c`. This is stronger and more relevant than another plot of first-fall degree.

The Kummer group-algebra description suggests a structured ansatz: index tower monomials by `z^k`, stratify by popcount, and allow the functional to depend on the top `c` carry levels where multiplication wraps modulo `z^(2^t)-1`. The one-summand and point-specialization maps are unsuitable because the existing proof shows they collapse to constant bounds. A useful search must retain at least the two-dimensional Kummer curve.

If such certificates persist for growing `c(t)`, they give an exact closure- degree lower bound. A matrix-width claim still needs a separate argument: degree alone does not lower-bound all representations or sparse work. The certificate verifier should therefore report the actual normal-form column support, row support, rank, nullity, and peak frontier. The existing `N=20` 240–380 GB estimate remains a secondary confirmation cell, not the proof programme.

5. Wagner bucket variance

The exact missing quantity is not an informal bucket imbalance. For bucket weights `q_c`, global agreement rate `delta`, and target-bucket rates `pi_c`,

```
W = sum_c q_c * (pi_c/delta - 1)^2.
```

It yields the checked one-sided bound `max_c(pi_c)/delta <= 1 + sqrt(W/min_c(q_c))`. When `q_c=Theta(1/M)` and `M*delta=1+o(1)`, the Wagner penalty is bounded by `G <= 1+sqrt(MW)(1+o(1))`. The proposed fit is `W=Theta(M^a*p^(-1/2))`; the proof-repair direction needs `a<1/2` at `M=p^(1/3)`.

Both outcomes must remain symmetric in interpretation. Small `W` supports a proof attempt that retires the claimed filtering gain. Large/nondecaying `W` breaks that repair and is only an attacker-favourable signal, not an attack by itself. The positive control is the discrete-log interval pullback, which must produce large `W`; SHA/matched-marginal shuffles are nulls. Exact whole-group anchors at the two smallest primes and jackknife bias checks are mandatory.

The existing implementation lineage is legacy Python. Repository policy and the scientific contract require a native Rust producer, frozen 8-prime ladder, `M` grid, `10^7`-pair budget, checkpoint state, estimator replay, and EKS Spot admission before execution. No result was manufactured from the old first- moment data.

6. Xedni selected-lift/minimal-discriminant bridge

The quantifier order must be fixed before examples. A valid target has the form: for one explicit selector `S`, one public curve/lift family, and every finite-field input in that family, `S` terminates and its selected rational points satisfy a uniform height bound against the global minimal discriminant. The selector must also preserve the reduction map and prime-subgroup sampling.

Two easy substitutions are invalid:

- bounding every rational multiple is false on a positive-rank curve by the existing canonical-height theorem; and
- bounding against the discriminant of whichever integral model the algorithm happens to receive is not intrinsic, because admissible scaling multiplies the model discriminant by a twelfth power while leaving an isomorphic curve.

Therefore the first selector experiment must normalize to a global minimal model, not merely a short or locally minimal model. A concrete initial selector is allowed to be partial: enumerate rational lifts in increasing certified naive-height boxes and select the lexicographically first lift that reduces to the input point. The research question is then explicit: prove termination and a uniform selected-height/minimal-discriminant ratio, or preserve the first family for which the ratio is unbounded. Without termination, global minimality, and good-reduction correspondence, this is only a search rule.

Even a successful selected-height theorem would leave Lang/direct gap, Mordell–Weil rank and torsion fibres, useful nonzero denominators, cofactor, and adaptive-hazard composition as separate graph nodes.

Validation performed

The tensor module was built in Rust 1.90 with the repository lockfile; four targeted tests passed. The default boundary CLI reported every numeric guard true and `live_host_representable=false`. The arity-three native crate was built with its lockfile; all three new trail tests passed. `git diff --check` is part of the final repository validation.

No EKS scientific job was launched. Runtime/AWS readiness does not override the yield custody hold, missing Wagner native producer, incomplete arity-three propagator, PKM proof target, or xedni selector theorem.